

Modern Workflow in Geology

Process Structural and Petrological Data Efficiently

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Welcome

Take home ready-made notebooks

Jupyter notebooks into which you simply “pour” your own Excel tables — structural measurements or microprobe data — and get stereograms, recalculations and charts.

Assumed: basic Python, Jupyter Lab, and how a pandas DataFrame works.

Tools: apsg (structural geology) and petropandas (petrology), both built on the scientific Python stack.

Today's Schedule

Time	Block
09:00–09:30	Introduction and preparation of the working environment
09:30–11:00	Block 1: Structural geology with apsg
11:00–11:15	<i>Coffee break</i>
11:15–12:30	Block 2: Advanced structural and tensor analysis
12:30–13:30	<i>Lunch break</i>
13:30–15:00	Block 3: Processing petrological data with petropandas
15:00–15:15	<i>Coffee break</i>
15:15–16:30	Block 4: Bring your own data (BYOD)

Scientific Python

The Scientific Python Stack

numpy	arrays, linear algebra
pandas	tables (DataFrame), Excel/CSV I/O
matplotlib	plotting, PDF/SVG export
scipy	statistics, optimisation
jupyterlab	notebooks in the browser
apsg	orientation data, tensors
petropandas	mineral chemistry

Why bother?

- Every step is written down and re-runnable
- Raw data are never edited by hand
- 10 or 10 000 measurements — same code
- Figures regenerate when data change
- Free, open, and shared with the whole community

Installation

Install with uv

1. Install uv (one-time; a fast Python project manager):

```
# Linux / macOS
curl -LsSf https://astral.sh/uv/install.sh | sh
# Windows (PowerShell)
powershell -ExecutionPolicy Bypass -c "irm https://astral.sh/uv/install.ps1 | iex"
```

2a. Use the workshop materials (recommended — exact tested versions):

```
cd UPSC2026_workshop # cloned or unzipped (...-main) materials
uv sync
```

2b. ...or start your own project from scratch:

```
uv init --bare --python 3.14 myproject && cd myproject
uv add "apsg[lab]" "petropandas[lab]"
```

3. Start Jupyter Lab: `uv run jupyter lab`

Check the Installation

Open a new notebook and run:

```
import apsg, petropandas
print(aps.__version__, petropandas.__version__) # 2.0.5 0.2.3

from apsg import *
quicknet(fol(120, 40), lin(210, 30)) # a stereonet should appear

from petropandas import pd, Grt
from petropandas.data import minerals
minerals.oxides.select("Garnet", on="Mineral").mineral.end_members(Grt).head()
```

- Notebooks live in notebooks/, example data in notebooks/data/, your results go to notebooks/output/
- Problems? Raise your hand now — everything later builds on this

Two Philosophies

```
from apsg import *

f = fol(250, 40) # a plane
l = f ** fol(160, 30) # intersection line
S = folset.random_fisher(
    position=f, kappa=20) # a set of planes
S.ortensor().eigenlins() # principal axes

F = defgrad.from_comp(xx=2, zz=0.5)
S.transform(F) # deform the set

s = StereoNet()
s.great_circle(S); s.line(l); s.show()
```

- Data are **objects that know what they are**: a `lin` is not a `fol`, a fault has a sense
- Operators have geological meaning: `**` intersection, `@` tensor product
- Sets carry statistics; tensors transform anything
- Stereonets understand every object type

petropandas: DataFrame Integration

```
from petropandas import pd, Grt

df = pd.read_excel("epma_export.xlsx")
df.oxides() # oxide columns
df.moles() # molar proportions
df.cations(n_oxygens=12) # APFU
df.mineral.end_members(Grt)
df.mineral.check_stoichiometry(Grt)
```

- No new data structure — your table **stays a DataFrame**
- Functionality hangs off **accessors**: `df.oxides`, `df.moles`, `df.cations`, `df.mineral`, `df.bulk`
- All of pandas still works: filtering, grouping, `to_excel`

Bridge

apsg also has DataFrame accessors: `df.fol()`, `df.lin()`, `df.fault()` turn table columns into apsg sets.

Notebook	Content
01_apsg_basics	Block 1: features, stereonet, statistics, fold axes
02_apsg_advanced	Block 2: faults, strain, progressive deformation, stress inversion
03_petro_pandas	Block 3: EPMA export → formulas, end-members, QC, charts
04a_byod_structural	Block 4 template: your field data → stereonet + statistics
04b_byod_epma	Block 4 template: your microprobe data → tables + charts

Each teaching notebook ends with a hands-on exercise and a worked solution. Example data: [notebooks/data/](#).